

MLFF Architecture Revision 73

SIZE-HALVE2 fixed-eight qualified-only 3/10/30 funnel

mdstats project

2026-08-16

Revision 73

Revision 73 implements **SIZE-HALVE2** as a separate pre-migration authority. It freezes exactly eight possible target sizes from 128 through 16,384, admits only independently hard-qualified MV sizes to training, requires at least four qualifiers, and implements the exact $q \rightarrow \min(q, 4) \rightarrow 2 \rightarrow 1$ 3/10/30 evidence transitions.

The authority inherits **PERF-P2R** exact continuation: epoch-10 evidence authenticates epoch-3 checkpoint/optimizer/RNG state, epoch-30 evidence authenticates epoch-10 state, and all candidates share foundation, evaluation-role, **TRAIN2**-policy, training-run, and schedule identity, exact normalized schedule progress, and increasing exposure. Largest-boundary tie protection remains limited to early practical-equivalence bands; final selection prefers the smaller equivalent size and reports **nonconverged_at_fixed_ceiling** when the largest qualified boundary remains materially superior.

Campaign preparation persists **size_halfve2_plan** after **MVQUAL1** and includes it in restart receipts. This gate is intentionally pre-migration: revision-64 **TARGET-DATA2C** v4, **TARGET-DATA2D** v2, **DATA8** membership, and **CuEq** policy remain unchanged.

Release: mdstats 0.20.206a0

Dependency-graph schema: 55

Next gate: **SIZE-FIDELITY2**.